Unit-1: Machine Leaning

1. Introduction to Machine Learning

Machine Learning is a field of study that gives computers the ability to learn without being explicitly programmed. In traditional programming, a human programmer writes detailed rules (If/Then logic) for every possible scenario. In Machine Learning, we provide the computer with data and an algorithm, and the computer discovers the patterns or rules itself.

Tom Mitchell (1997) formalized the learning process into three distinct components:

- **Experience( )**
- **Task( )**
- **Performance( ).**

Mitchell stated that "**A computer program is said to learn from experience with respect to some class of tasks and performance measure, if its performance at tasks in, as measured by, improves with experience** ".

This definition is crucial for students to master because it provides a framework for identifying whether a system is truly "learning" or just following a static script. We can apply this framework to various real-world scenarios to test its validity.

Table: Machine Learning framework Real-world Scenarios

Learning Scenario	Task ( )	Experience ( )	Performance Measure ( )
Email Spam Filter	Classifying emails as 'Spam' or 'Not Spam'	Observing users mark emails as 'Spam' or 'Not Spam'	The percentage of emails correctly classified
Autonomous Driving	Driving on a public highway	Sequences of images and steering commands from human drivers	Average distance travelled without human intervention
Handwriting Recognition	Recognizing handwritten text from images	Database of handwritten words with known labels	The accuracy of word identification
Chess Engine	Playing the game of Chess	Playing practice games against itself or others	The percentage of games won against opponents

Real World Example: *Imagine you want to teach a computer to recognize a "cat." In traditional programming, you would have to describe exactly what a cat looks like: whiskers, two ears, four legs. But cats come in many colors and positions. In Machine Learning, you show the computer 10,000 photos of cats and say, "These are cats," and 10,000 photos of non-cats and say, "These are not." The computer learns the visual features that make a cat a cat.*

Why Machine Learning?

- ***Handling Large Data***: Humans cannot process millions of data points, but machines can in seconds.
- ***Adaptability***: ML models can update themselves as new data arrives, unlike rigid software.
- ***Discovering Hidden Insights***: Machines can find connections in data that are too subtle for humans to notice.

2. Types of Machine Learning Models

Machine Learning is generally categorized into three main types based on how they learn from the input data. The three primary paradigms are Supervised Learning, Unsupervised Learning, and Reinforcement Learning.

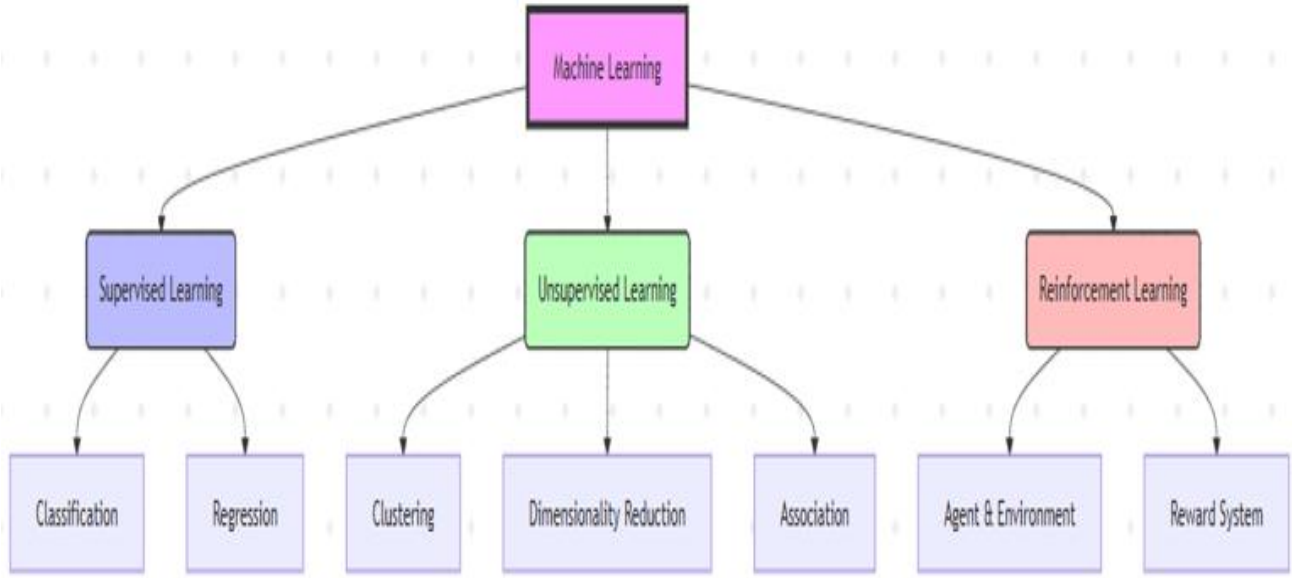

Fig: Types of Machine Learning Models

A. Supervised Learning

This is the most common type. Here, the model is trained on a "labeled" dataset. This means for every input piece of data, we also provide the "correct answer" or "target label." It is like a student learning with a teacher who corrects their homework.

Supervised learning problems are further divided into two sub-categories based on the nature of the target variable:

Regression: Regression is used when the output variable is a continuous numerical value. The goal is to predict a quantity.

Real-World Example: **House Price Prediction.**

- ***Input ( )***: Square footage, number of bedrooms, zip code, age of the house.
- ***Output ( )***: Price in Rupees (e.g., 4,50,000).
- ***Logic:*** The model learns a mathematical relationship (e.g., a line or curve) that correlates features like size with price. Since price can be any number within a range, this is a regression task.

Other Examples: **Predicting stock prices, estimating temperature, forecasting sales revenue.**

Classification: Classification is used when the output variable is a categorical discrete value. The goal is to predict a label or group.

Real-World Example: **Medical Diagnosis.**

- ***Input ():*** Patient's blood pressure, cholesterol levels, age, BMI.
- ***Output ():*** Disease status (e.g., "Has Heart Disease" or "Healthy").
- ***Logic:*** The model learns a decision boundary that separates the data points into distinct classes. Even though the inputs are numbers, the output is a category (Yes/No), making this a classification task.

Other Examples: **Spam detection (Spam/Not Spam), Image recognition (Cat/Dog), Sentiment analysis (Positive/Negative).**

B. Unsupervised Learning

Unsupervised Learning involves training a machine on data that has no labels. The system is provided with input data but no corresponding output. The goal is not to predict a specific answer but to discover hidden structures, patterns, or relationships within the data.

This is analogous to a child playing with a bucket of mixed Lego blocks. Without being told what to do, the child might spontaneously group the blocks by color (reds, blues) or by size (large, small). The child is learning the inherent structure of the blocks without a "teacher" providing correct answers.

Unsupervised learning problems are further divided into three sub-categories based on the nature of the input variable:

Clustering: Clustering is the task of grouping a set of objects in such a way that objects in the same group (called a cluster) are more similar to each other than to those in other groups.

Real-World Example: **Customer Segmentation**.

- A retail company has millions of transaction records but does not know who its "best" customers are.
- By feeding this data into a clustering algorithm, the system might discover three distinct groups: "Budget Shoppers" (buy only on sale), "Brand Loyalists" (buy frequent, high-value items), and "Seasonal Shoppers" (buy only during holidays).
- The marketing team can then target each group with a different strategy.

Dimensionality Reduction: Dimensionality Reduction is the process of reducing the number of random variables under consideration by obtaining a set of principal variables. It is often used as a pre-processing step to simplify data.

Real-World Example: **Image Compression.**

- A high-resolution image consists of millions of pixels (features). Many of these pixels are redundant (e.g., a large patch of blue sky).
- Dimensionality reduction techniques like Principal Component Analysis (PCA) can compress this data into fewer features while retaining the essential information (the edges and shapes) required to recognize the image.

Association Rule Learning: This technique discovers interesting relations between variables in large databases.

Real-World Example: **Market Basket Analysis.**

- Retailers analyze purchase history to find items that are frequently bought together.
- "If a customer buys Bread and Milk, they are 80% likely to buy Eggs."
- This insight drives store layout and cross-selling strategies.

C. Reinforcement Learning

Reinforcement Learning (RL) differs significantly from both supervised and unsupervised learning. It is a behavioral learning model where an agent learns to make decisions by performing actions in an environment and receiving feedback in the form of rewards or penalties. If the action is good, it gets a "Reward." If the action is bad, it gets a "Penalty." The goal is to maximize the total reward over time.

Key Components:

Agent: The learner or decision-maker (e.g., a robot).

Environment: The world the agent interacts with (e.g., a maze).

Action: A move the agent makes (e.g., move left).

State: The current situation of the agent (e.g., position in the maze).

Reward: Numerical feedback (e.g., +10 for finding cheese, -1 for hitting a wall).

Real-World Example: **Robotic Navigation.**

- A robot vacuum cleaner learning to navigate a room.
- If it cleans dirt, it gets a positive reward. If it bumps into a table, it gets a negative penalty.
- Over time, the robot learns a "policy" (a strategy) to clean the room efficiently while avoiding obstacles.

Table: Comparison of Machine Learning Types

Feature	Supervised Learning	Unsupervised Learning	Reinforcement Learning
Input Data	Labeled Data (Input + Output)	Unlabeled Data (Input only)	States and Actions (No fixed dataset)
Goal	Predict outcomes/labels	Discover hidden patterns/structures	Maximize cumulative reward
Feedback Mechanism	Direct and immediate (Correct/Incorrect)	No feedback; internal evaluation	Delayed feedback (Reward/Penalty)
Complexity	Moderate; maps Input to Output	High; finds structure in chaos	Very High; interacts with dynamic
Key Algorithms	Linear Regression, Decision Trees, SVM	K-Means, PCA, Apriori	Q-Learning, Deep Q-Networks (DQN)

3. Applications of Machine Learning

Machine Learning has transitioned from academic theory to practical ubiquity. It powers the core functionality of many modern technologies.

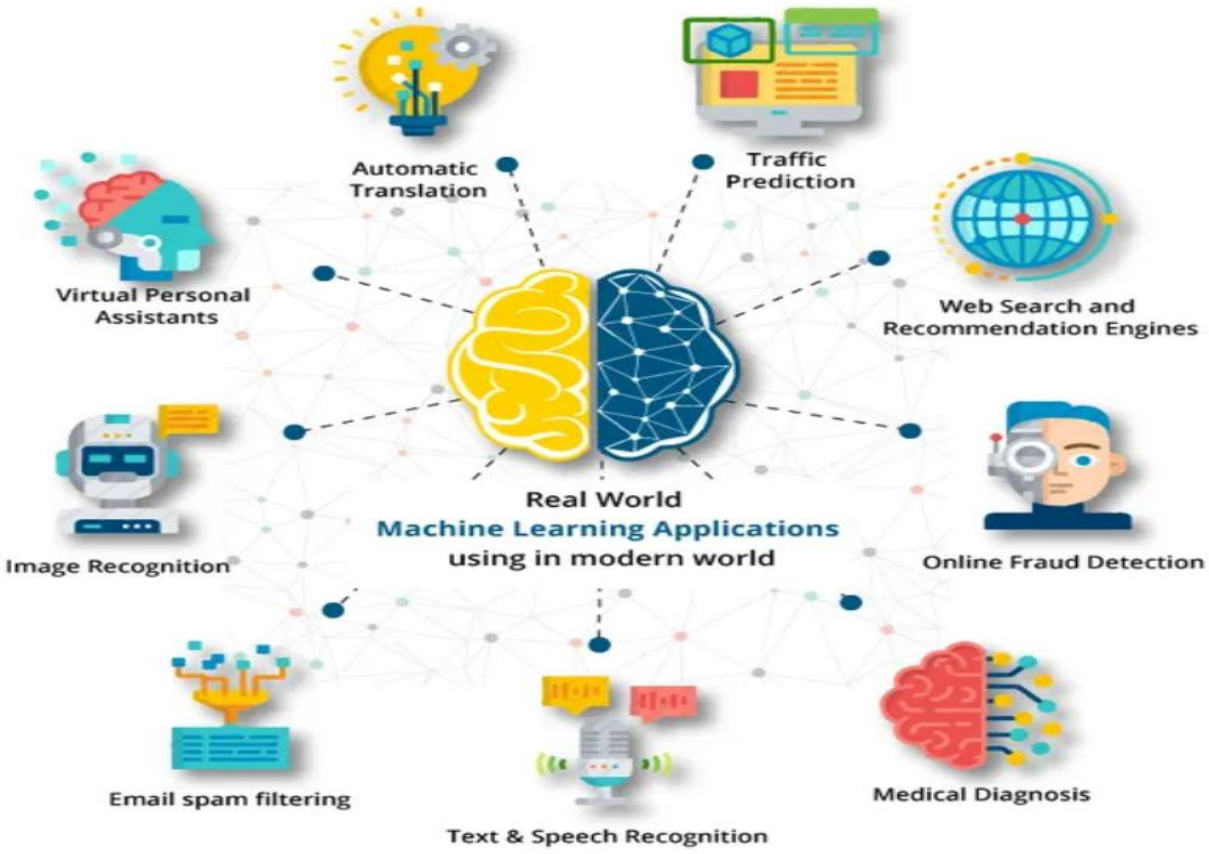

Fig: Applications of Machine Learning

Healthcare: The medical field generates vast amounts of complex data, making it an ideal candidate for ML applications.

- ***Medical Imaging***: ML algorithms, particularly Deep Learning models, can analyze X-rays, MRIs, and CT scans to detect anomalies like tumors or fractures with accuracy often surpassing human radiologists. For instance, detecting diabetic retinopathy from retinal scans.
- ***Precision Medicine***: By analyzing a patient's genetic profile and medical history, ML models can predict how a patient will respond to specific drug treatments, allowing for personalized care plans rather than a "one size fits all" approach.
- ***Predictive Analytics***: Hospitals use ML to predict patient inflow, identifying peak times to optimize staffing and resource allocation. It is also used to predict septic shock in ICU patients' hours before clinical symptoms appear.

Finance and Banking: Financial institutions were early adopters of ML due to the high value of predictive accuracy in money management.

- ***Fraud Detection:*** This is a classic anomaly detection problem. ML systems monitor millions of transactions in real-time. If a credit card is used for a high-value purchase in a country where the owner does not live, the system flags the transaction as an outlier and blocks it instantly.
- ***Algorithmic Trading***: High-frequency trading firms use ML to analyze market news, social media sentiment, and historical price trends to execute trades in milliseconds, capitalizing on minute market inefficiencies.
- ***Credit Scoring***: Traditional credit scoring relies on static rules. ML models incorporate alternative data sources (e.g., utility bill payments, rental history) to assess the creditworthiness of individuals with thin credit files.

Retail and E-Commerce

- ***Recommender Systems:*** Platforms like Amazon and Netflix utilize Collaborative Filtering algorithms. These systems analyze a user's past behaviour (purchases, views) and compare it with similar users to recommend new products or movies. This drives a significant portion of revenue for these companies.
- ***Inventory Optimization***: ML models forecast demand for products based on factors like weather, local events, and historical sales data. This helps retailers stock the right amount of perishable goods, reducing waste and lost sales.

Manufacturing

- ***Predictive Maintenance***: Sensors on factory machines collect vibration, temperature, and sound data. ML models analyze this data to predict when a machine is likely to fail *before* it happens. This allows for maintenance to be scheduled proactively, preventing costly unplanned downtime.
- ***Quality Control:*** Computer vision systems inspect products on assembly lines at high speed, automatically rejecting items with scratches, dents, or other defects that might be missed by human inspectors.

Transportation and Logistics

- ***Autonomous Vehicles***: Self-driving cars (e.g., Tesla, Waymo) process streams of data from cameras, LiDAR, and radar. Supervised learning is used to recognize stop signs and pedestrians, while reinforcement learning helps the car make navigation decisions in complex traffic.

- ***Route Optimization***: Logistics companies like UPS and FedEx use ML to plan delivery routes that minimize fuel consumption and delivery time, factoring in real-time traffic and weather conditions.

4. Preparing to Model

Basic Types of Data: Data is the fuel for machine learning. To process it effectively, we must first understand its type. Data is broadly classified into Quantitative (Numerical) and Qualitative (Categorical).

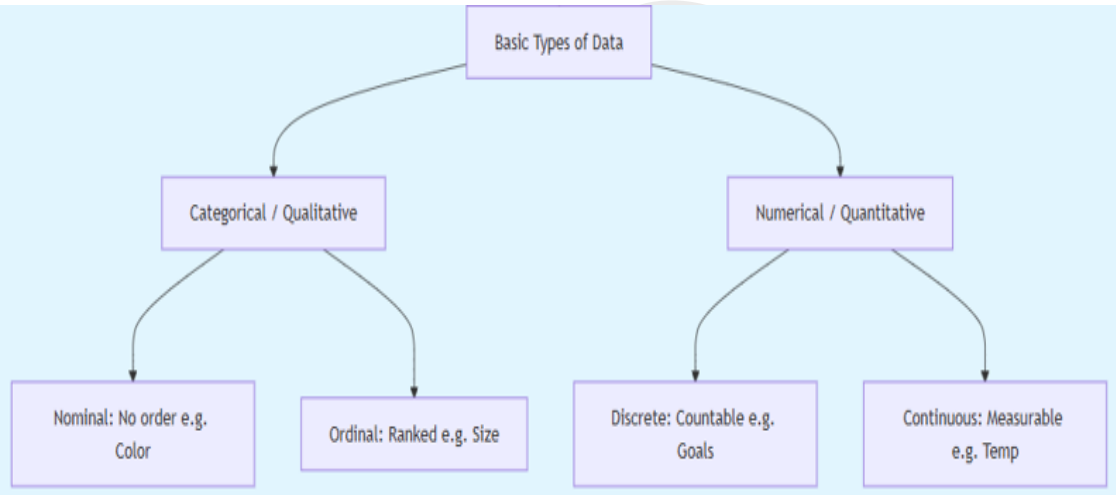

Fig: Machine Learning Basic Data Types

Quantitative (Numerical) Data: This data represents quantities and amounts. It can be measured and ordered.

Discrete Data: Data that can only take specific, countable values. It cannot be subdivided.

Example: The number of students in a class (you cannot have 10.5 students), the number of cars in a parking lot, the result of rolling a dice (1, 2, 3, 4, 5, 6).

Continuous Data: Data that can take any value within a range. It can be infinitely subdivided.

Example: Height (175.5 cm), Weight (65.23 kg), Temperature (98.6°F), Time taken to run a race (10.45 seconds).

Qualitative (Categorical) Data: This data represents characteristics or qualities. It describes "what" rather than "how much."

Nominal Data: Categories with no intrinsic order or ranking. The labels are just names.

***Example**:* Colors (Red, Blue, Green), Gender (Male, Female), Country (India, USA, Japan). It makes no sense to say "Red > Blue" or calculate the average of "India" and "USA".

***Ordinal Data**:* Categories that have a clear, natural order, but the intervals between them are not necessarily equal.

***Example**:* Customer Satisfaction (Unhappy, Neutral, Happy), Education Level (High School, Bachelor's, Master's), Shirt Size (S, M, L). We know that Master's is "higher" than High School, but we cannot quantify the "distance" between them mathematically.

Levels of Measurement: It is also useful to classify data by its level of measurement, which dictates the statistical operations permissible:

Interval Data: Ordered data where the difference between two values is meaningful, but there is no true zero point.

***Example**:* Temperature in Celsius. The difference between 20°C and 30°C is the same as 30°C and 40°C, but 0°C does not mean "no temperature."

Ratio Data: Ordered data with meaningful differences and a true zero point.

***Example**:* Weight. 0 kg means "no weight," and 20 kg is exactly twice as heavy as 10 kg.

5. Exploring Structure

This is often called Exploratory Data Analysis (EDA). It involves checking the dimensions of the data (number of rows and columns) and understanding the distribution of each feature. We look for the "**Mean**" (Average), "**Median**" (Middle value), and "**Standard Deviation**" (How much the data varies).

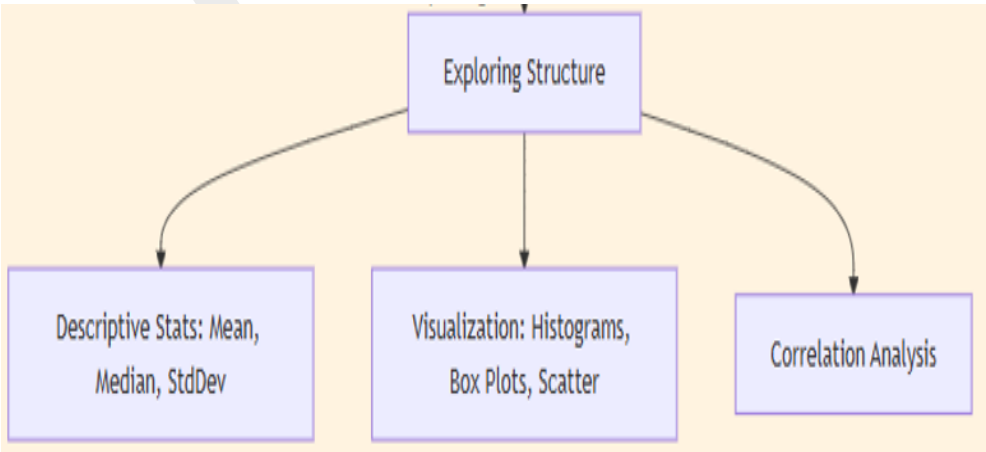

Fig: Exploring Structure

Descriptive Statistics

- ***Measures of Central Tendency***: Mean (average), Median (middle value), Mode (most frequent value). For skewed data (like salaries), the Median is often more representative than the Mean.
- ***Measures of Dispersion***: Range, Variance, Standard Deviation. These tell us how spread out the data is. A high standard deviation means the data is very volatile.

Data Visualization

Humans are visual creatures. We spot patterns in charts that we miss in spreadsheets.

Histogram: A bar chart that groups numerical data into "bins" (ranges).

Purpose: To see the distribution of data. Is it symmetrical (Bell Curve)? Is it skewed to the left or right?

Example: Plotting the ages of a population. If the bars are highest on the left, it is a young population (Right Skewed).

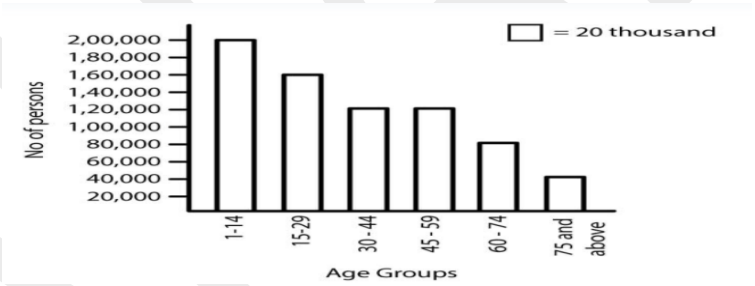

Fig: Histogram

Box Plot (Box-and-Whisker Plot): A standardized way of displaying the distribution of data based on a five-number summary: Minimum, First Quartile (Q1), Median, Third Quartile (Q3), and Maximum.

Purpose: Ideally suited for detecting outliers. Any data point that falls outside the "whiskers" (typically 1.5 times the Interquartile Range) is considered an anomaly. It also allows for easy comparison between groups.

Example: Salary distribution of Males vs. Females.

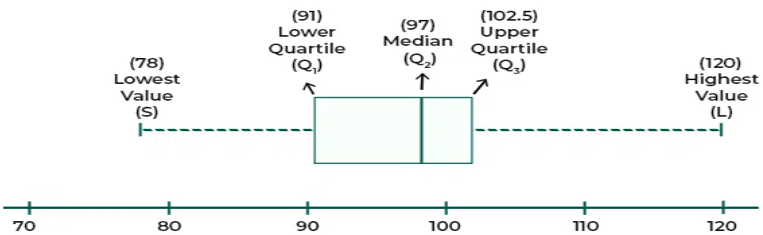

Fig: Box plot

Scatter Plot: A graph in which the values of two variables are plotted along two axes.

Purpose: To reveal relationships or correlations between two continuous variables.

Example: Plotting "Years of Experience" vs. "Salary." If the points drift upwards from left to right, there is a positive correlation.

Fig: Scatter plot

6. Data Quality, Remediation & Pre-processing

A common misconception is that Machine Learning is mostly about selecting algorithms. In reality, 70-80% of a data scientist's time is spent on data preparation. The performance of a model is bounded by the quality of the data it is trained on; a principle known as "Garbage In, Garbage Out."

Data Quality and Remediation: Data in the real world is "dirty." "Garbage In, Garbage Out" is a famous saying in Machine Learning: if you give the model bad data, it will give you bad results.

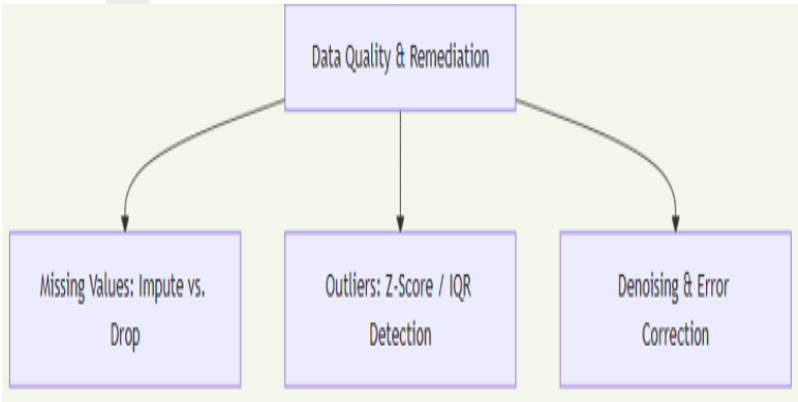

Fig: Data Quality & Remediation

Missing Values: Data can be missing for various reasons: a sensor malfunction, a user refusing to answer a survey question, or a file corruption.

Remediation Strategies:

- ***Deletion***: If a row (record) has missing values, simply remove it. This is safe if the dataset is huge and the missing rows are few (e.g., < 5%). However, if data is limited, this causes loss of information.
- ***Imputation (Mean/Median)***: Fill the missing value with the average (Mean) or middle value (Median) of the column. Median is preferred if the data has outliers.
- ***Imputation (Mode):*** For categorical data (e.g., Gender), fill the missing value with the most frequent category.
- ***Prediction***: Use a machine learning algorithm (like KNN) to predict the missing value based on other features.

Outliers: Outliers are data points that differ significantly from other observations. They can be caused by measurement errors (e.g., Human Height = 10 feet) or genuine variability (e.g., Bill Gates' income).

Remediation Strategies:

- ***Removal:*** If the outlier is clearly an error, delete it.
- ***Capping (Winsorizing):*** Set a threshold (e.g., 99th percentile). Any value above this is replaced by the threshold value.
- ***Transformation:*** Apply a Log Transformation to compress the scale of the data, pulling outliers closer to the mean.

Duplicate Data: Duplicate records can bias the model, causing it to "over-learn" specific patterns.

Remediation: Identify and remove duplicate rows based on a unique identifier or a combination of all columns.

Data Pre-Processing: This step involves transforming raw data into a clean, digital format that an algorithm can process.

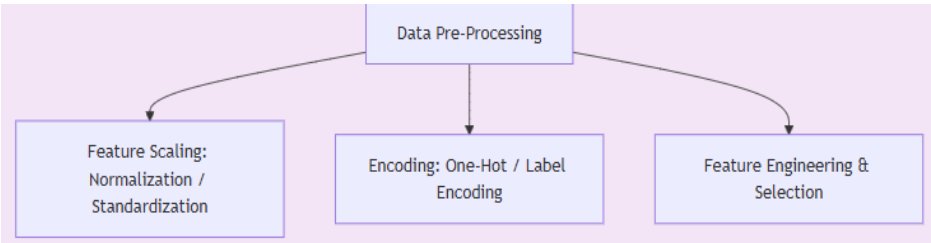

Fig: Data Pre-processing

- ***Normalization/Scaling***: If one feature is "Salary" (50,000 to 100,000) and another is "Age" (18 to 80), the large salary numbers will dominate. We scale them both to be between 0 and 1.
- ***Data Integration:*** Merging data from multiple sources (SQL databases, CSV files, APIs) into a unified dataset.
- ***Data Reduction***: Reducing the volume of data while maintaining its integrity (e.g., removing irrelevant columns).
- ***Data Transformation***: Normalization and Encoding (covered in detail in the Feature Engineering section). Machines don't understand "Red" or "Blue." We convert them to 0s and 1s.
- ***Discretization:*** Converting continuous attributes into discrete intervals. For example, converting "Age" (0-100) into buckets: "Child" (0-12), "Teen" (13-19), "Adult" (20-60), "Senior" (60+).

7. Modelling and Evaluation

Once the data is prepared, we move to the core of machine learning: building and testing the model.

Selecting a Model: There is no single "best" algorithm. The choice depends on the problem and the data. Not every algorithm works for every problem. Selection depends on the size of the data, the type of output (number or category), and the complexity of the patterns.

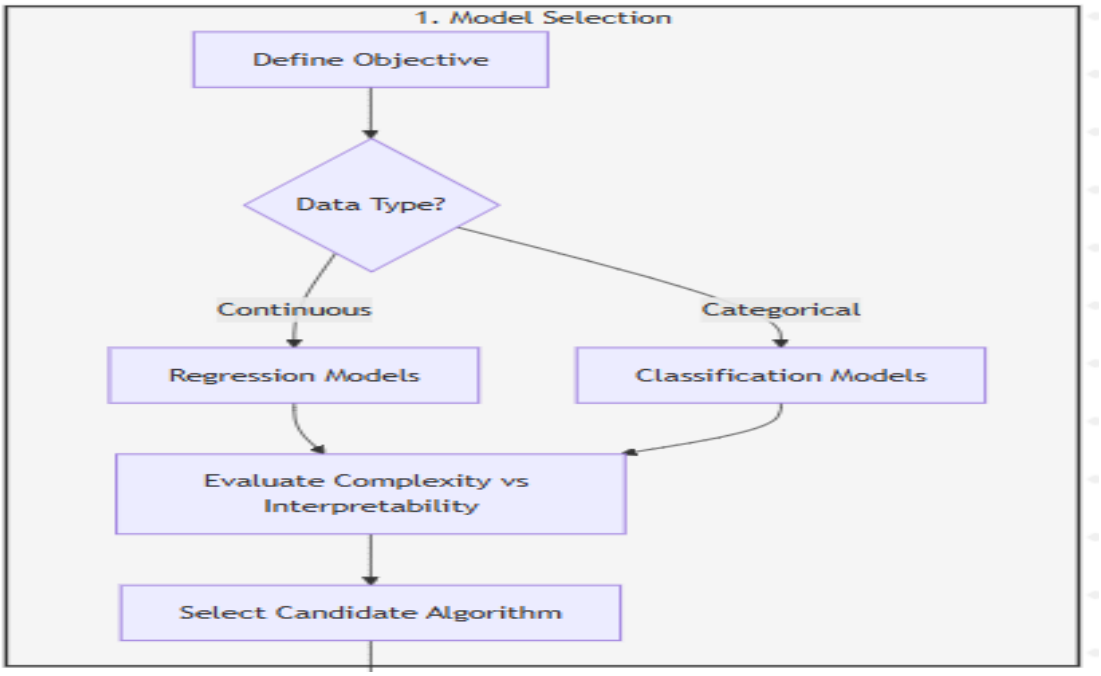

Fig: Model Selection

Predictive Models (Supervised): If the goal is to predict a future outcome.

- Use **Linear Regression** if the target is a number.
- Use **Logistic Regression or Decision Trees** if the target is a category.

Descriptive Models (Unsupervised): If the goal is to understand patterns.

- Use **K-Means or Hierarchical Clustering**.

The Bias-Variance Trade-off: This is the central problem in supervised learning. It helps us understand the errors in our models.

Bias (Underfitting): The error introduced by approximating a real-world problem with a too-simple model. A model with high bias pays very little attention to the training data and oversimplifies the model.

- ***Analogy***: A student who doesn't study and assumes all answers are "C". They will consistently get a low score (high error) regardless of the exam questions.

Variance (Overfitting): The error introduced by the model's sensitivity to small fluctuations in the training set. A model with high variance pays too much attention to the training data, capturing random noise rather than the intended outputs.

- ***Analogy:*** A student who memorizes the textbook word-for-word but doesn't understand the concepts. They will ace the practice test (training data) but fail the final exam (test data) because the questions are slightly different.

The Goal: We want a "Goldilocks" model: low bias and low variance.

Training a Model: Training involves feeding data to the algorithm so it can adjust its internal parameters (weights). However, we must ensure the model generalizes to new data. We usually split our data into two parts: Training Set (80%) and Testing Set (20%). The model learns from the Training set. We then use the Testing set to see if the model can predict answers for data, it has never seen before.

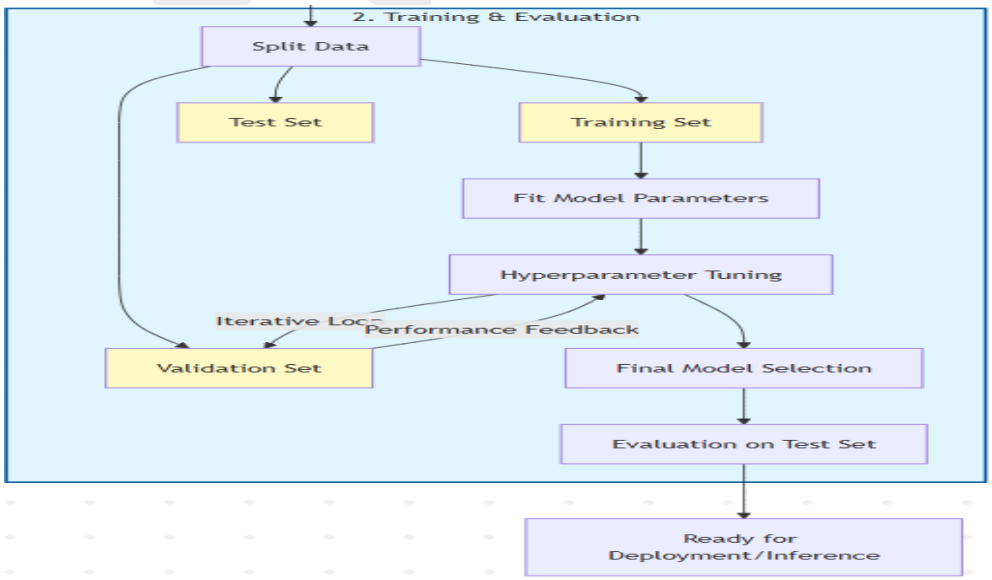

Fig: Model Training and Evaluation

<u>The Holdout Method</u>: This is the simplest evaluation technique.

1. Shuffle the dataset.
2. Split it into two parts: Training Set (e.g., 70%) and Test Set (e.g., 30%).
3. Train the model on the 70%.
4. Test the model on the hidden 30% to calculate accuracy.

***Drawback**:* If the 30% test set happens to contain the "hardest" examples, the model will look bad, even if it is good.

<u>K-Fold Cross-Validation</u>: This is a more robust statistical method.

1. Divide the dataset into equal subsets (folds). Common choices are or.
2. Train the model on folds and test it on the remaining 1-fold.
3. Repeat this process times, ensuring every fold serves as the test set exactly once.
4. The final score is the average of the scores.

***Advantage**:* It uses all data for both training and testing, providing a more reliable estimate of performance.

<u>Bootstrapping</u>: This involves random sampling with replacement. We create multiple training datasets by picking rows randomly from the original data, allowing the same row to be picked multiple times. This is often used in Ensemble methods like Random Forest.

<u>Model Representation and Interpretability</u>: Model representation is the mathematical way the model stores its knowledge. Interpretability is how easily a human can understand "why" a model made a specific prediction.

<u>Model Representation</u>: How the machine stores what it has learned.

- ***Linear Regression**:* Stored as a set of coefficients () and an intercept (). The model is the equation.
- ***Decision Tree**:* Stored as a set of If-Then rules in a tree structure.
- ***Neural Network**:* Stored as a matrix of numerical weights connecting layers of neurons.

<u>Interpretability:</u> The degree to which a human can understand the cause of a decision.

- White Box Models: Highly interpretable. (e.g., Decision Trees). We can trace the path: "The loan was denied because Income < 50k AND Credit Score < 600."
- Black Box Models: Low interpretability. (e.g., Deep Learning). The model says "Loan Denied," but the reason is buried in millions of mathematical operations. Techniques like LIME (Local Interpretable Model-agnostic Explanations) are used to try and explain these black boxes post-hoc.

Example: *A "Decision Tree" is highly interpretable because you can follow the branches. A "Deep Neural Network" is often a "Black Box" because even scientists don't exactly know how it calculated the final result.*

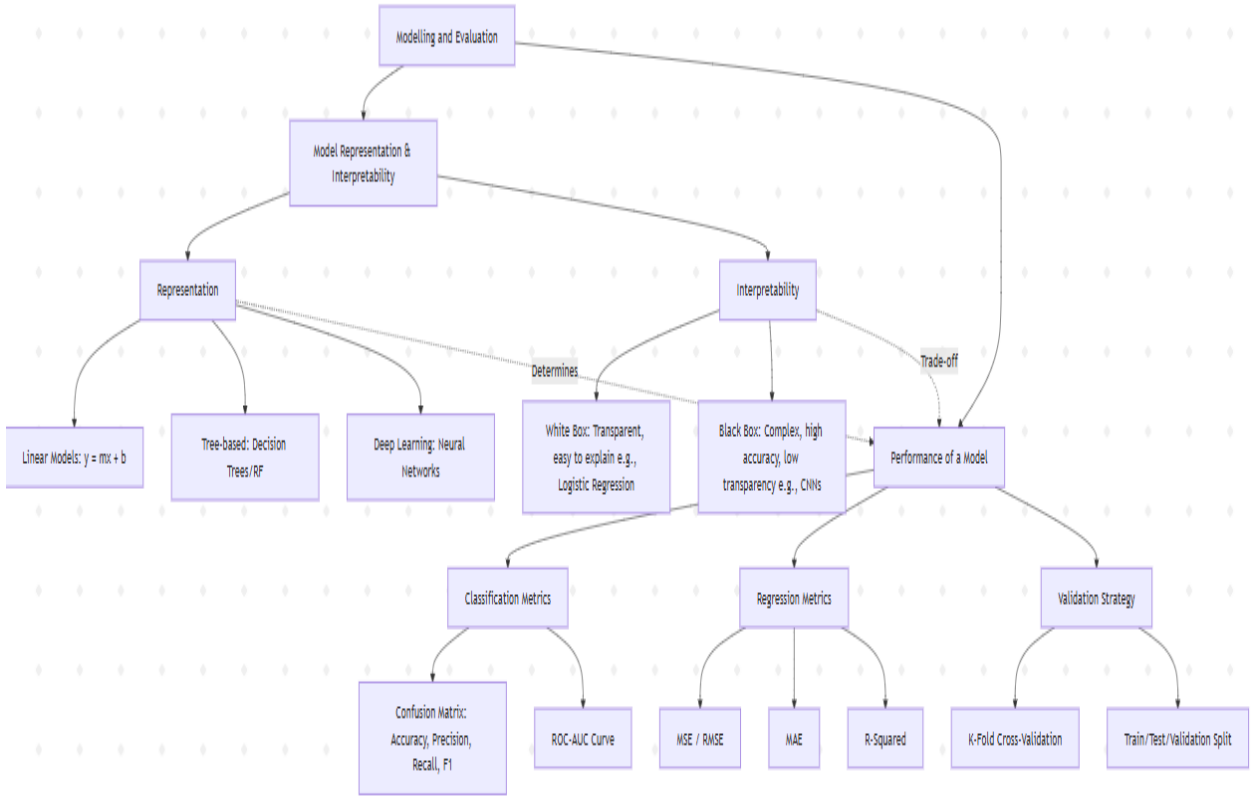

Fig: Representation of a Model

Performance of a Model

We cannot rely on "feeling" if a model is good. We need mathematical metrics.

Classification Metrics

Confusion Matrix: A table that breaks down predictions.

- ***True Positive (TP)***: Predicted YES, Actual YES.
- ***True Negative (TN):*** Predicted NO, Actual NO.
- ***False Positive (FP):*** Predicted YES, Actual NO (Type I Error). *Example: You are healthy, but the test says you have cancer.*
- ***False Negative (FN):*** Predicted NO, Actual YES (Type II Error). *Example: You have cancer, but the test says you are healthy.*

Accuracy: The most intuitive metric. That is percentage of total correct predictions

- *Trap:* If 99% of transactions are legitimate and 1% are fraud, a model that simply predicts "Legitimate" for *everything* will have 99% accuracy but 0% fraud detection. In such imbalanced cases, accuracy is useless.

Precision: Quality of positive predictions. That is out of all "Yes" predictions, how many were actually "Yes"?

- *Use Case:* Spam detection. We want high precision because we don't want to accidentally delete a real email (False Positive).

Recall (Sensitivity): Quantity of positive predictions. That is out of all actual "Yes" cases, how many did the model catch?

- *Use Case:* Cancer detection. We want high recall because missing a cancer case (False Negative) is fatal.

F1 Score: The Harmonic Mean of Precision and Recall. That is a balance between Precision and Recall. It penalizes extreme values.

Regression Metrics: Since regression predicts numbers, we measure the "distance" between the prediction and reality.

- ***Mean Absolute Error (MAE):*** The average of the absolute errors. . It is robust to outliers.
- ***Mean Squared Error (MSE):*** The average of squared errors. . It punishes large errors heavily.
- ***Root Mean Squared Error (RMSE):*** The square root of MSE. It brings the unit back to the original scale (e.g., Dollars).

8. Feature Engineering

This is the "Art" part of Machine Learning. It involves using human intelligence to create better inputs for the computer. Feature Engineering is the process of using domain knowledge to extract features (characteristics, properties, attributes) from raw data. It is often considered the "secret sauce" of winning machine learning models.

Feature Transformation: Changing the existing data to make it more useful. Algorithms often require data to be in a specific format or scale.

For example, if you have a "Date of Birth," the model might learn better if you transform it into a new feature called "Age."

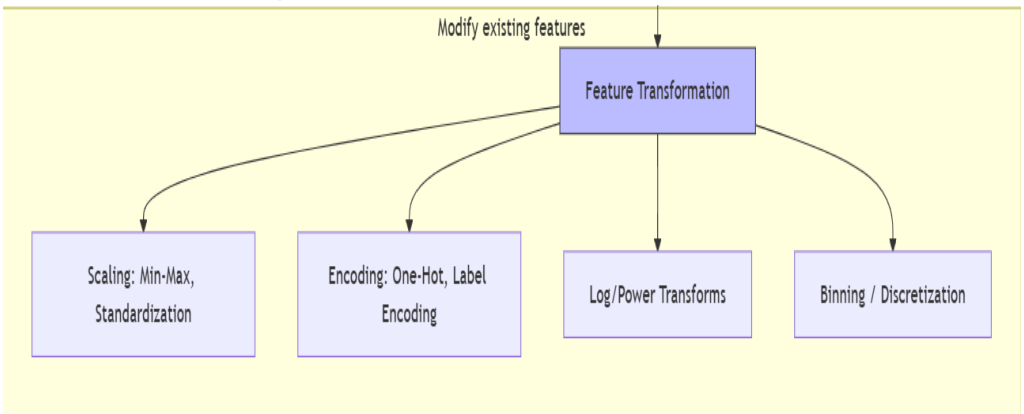

Fig: Future Transformation

Scaling (Normalization & Standardization): Many algorithms (like KNN and SVM) calculate the Euclidean Distance between data points. If features have different scales, the larger one will dominate.

Example:

- Feature A: Age (Range 0-100)
- Feature B: Salary (Range 20,000 - 200,000)
- A small change in Salary (e.g., 1000) looks huge compared to a change in Age (e.g., 50). The algorithm will essentially ignore Age.

Normalization (Min-Max Scaling): Rescales the feature to a fixed range, usually 0 to 1.

Standardization (Z-Score Scaling): Rescales the feature so it has a Mean of 0 and a Standard Deviation of 1.

Standardization is generally less affected by outliers than Normalization.

Encoding Categorical Data: Machine learning models are mathematical; they operate on numbers, not strings. We must convert text categories into numbers.

Label Encoding: Assigns a unique integer to each category.

Example: Apple=1, Banana=2, Cherry=3.

Problem: The model might misunderstand the data as ordinal, thinking Cherry (3) is "greater than" Apple (1), which is false. This is bad for Nominal data.

One-Hot Encoding: Creates a new binary column (dummy variable) for each category.

Example:

- Column Is_Apple: 1 if Apple, 0 otherwise.
- Column Is_Banana: 1 if Banana, 0 otherwise.

Benefit: It correctly represents nominal data without implying any order.

Feature Subset Selection: Providing too much information can confuse the model (this is called "The Curse of Dimensionality"). Feature selection is the process of picking only the most important features. If you are predicting house prices, the "Color of the Front Door" might not be important, so you remove it from the model to make it faster and more accurate.

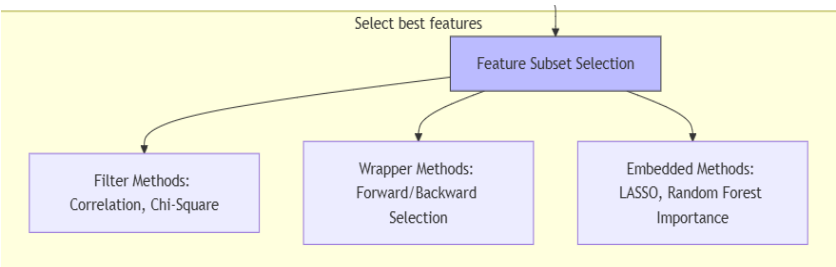

Fig: Feature Subset Selection

Modern datasets can have thousands of features. Using all of them can lead to slow training and overfitting (Curse of Dimensionality). We need to select the best subset.

Filter Methods: Filter methods select features based on statistical scores, independent of the machine learning algorithm. They are fast and done *before* modeling.

- Correlation: Remove features that are not correlated with the target variable.
- Variance Threshold: Remove features that have zero variance (i.e., the value is the same for all rows). If a column contains "Country: India" for every single row, it provides no information. Remove it.

Wrapper Methods: Wrapper methods evaluate subsets of variables which allows detecting the possible interactions between variables. They "wrap" the model training process.

Forward Selection:

i. Start with 0 features.
ii. Train the model with each feature individually. Pick the best one.
iii. Add a second feature that works best with the first.
iv. Repeat until performance stops improving.

Backward Elimination:

i. Start with all features.
ii. Train the model. Calculate the importance of each feature.
iii. Remove the least important feature.
iv. Repeat until performance drops significantly.

Recursive Feature Elimination (RFE): A greedy optimization algorithm which aims to find the best performing feature subset by repeatedly constructing models and removing the weakest feature.

Embedded Methods: These methods perform feature selection during the model training process itself.

- ***Lasso Regression (L1 Regularization):*** This algorithm penalizes complex models by shrinking the coefficients of less important features to exactly zero, effectively removing them.
- ***Tree-based Importances***: Algorithms like Random Forest calculate "Feature Importance" scores automatically, showing which features contributed most to the decisions.